Chemical Conversion Coatings (Alodine) – Manual Application

1. General

WARNING **THIS PROCEDURE USES MATERIALS THAT MAY BE DANGEROUS TO THE PERSONNEL'S HEALTH AND/OR THE ENVIRONMENT. SUFFICIENT PRECAUTIONS MUST BE TAKEN AT ANY STEP OF USE. IF NOT, THIS CAN CAUSE DAMAGE TO THE ENVIRONMENT AND DANGEROUS EFFECTS ON THE PERSONNEL'S HEALTH AND SAFETY.**

Manually apply chemical conversion coating solution only if application by immersion is not permitted due to the part's size or the assemblies' potential for solution entrapment. For touch-up of small surfaces (i.e. spotfacing, countersinks, repair areas etc.), it is recommended that Touch-N-Prep Alodine 1132 markers be used.

If assemblies contain materials other than aluminum or titanium, take extreme care when processing these assemblies because the chemical conversion coating solution may cause corrosion. It is recommended that material other than aluminum or titanium be masked off when processing using aluminum foil tape.

Manual application meets all requirements of MIL–DTL–5541 and covers two classes of chemical conversion coatings formed by the reaction of the chemical materials and the surfaces of aluminum and aluminum alloys:

- Class 1A coatings are corrosion protective films for parts, repair and touch-up applications and will improve paint adhesion for aluminum and aluminium alloys.
- Class 3 coatings are corrosion preventative films for electrical and electronic applications where low resistance contacts are required.

2. Materials and Equipment

- Alodine 1200S
- De-Ionized Water
- Scotch-Brite Pads
- Fine Abrasive Paper
- Suitable Brushes
- Polyethylene Container
- Clean Cotton Wipers
- Approved Safety Glasses
- Protective Rubber Gloves
- Deoxidine 624 (or Alumi Prep 33 or Turco W.O. 1)
- Approved Respirator
- Cellulose Sponge
- Touch-N-Prep Alodine 1132 Marker
- Aluminum Foil Tape, adhesive backed (3M #425)

3. Safety Precautions

A. Wear neoprene rubber gloves (refer to Para 2.), aprons, boots and approved chemical splash goggles while carrying out chemical conversion coating operations.

B. Ensure adequate ventilation is supplied to the areas where the chemical conversion coating process is being carried out. Avoid breathing the fumes or vapours during application.

C. Take care to prevent the powder from being distributed into the air. Airborne powder will cause a serious health hazard. Wear protective respiratory equipment when handling conversion coating materials.

D. Avoid skin contact with solutions. If skin contact occurs, wash the affected area immediately with large quantities of water. If irritation of the skin occurs, contact a First Aid attendant immediately.

E. Avoid eye contact with chemicals or solutions. If eye contact occurs, immediately flush eyes in a directed stream of water for at least 15 minutes while forcibly holding eyelids apart to ensure completed irrigation of all eye and lid tissue. Contact a First Aid attendant and a physician.

F. Avoid ingestion of chemicals or solutions. If ingestion occurs, do not induce vomiting. Drink large amounts of water or milk of magnesia, but never give anything by mouth to an unconscious person. Immediately contact a First Aid attendant and a physician.

4. Cleaning before Manual Application

WARNING **KEEP SOLVENT AWAY FROM FIRE AND OTHER SOURCES OF IGNITION. PROVIDE SUFFICIENT VENTILATION WHEN WORKING IN CONFINED SPACES. AVOID SKIN CONTACT WITH SOLVENT.**

Before manual chemical conversion coating as specified in Para 6. (except Type II IVD aluminum coated parts):

- Degrease as per Para 4.A
- If heavy oxides are present, mechanically clean according to Para 4.B
- Manually deoxidize according to Para 4.C only if using Alodine 1200S
- If Touch-up refer to Para 6.B, do not deoxidize

A. Degreasing and Manual Solvent Cleaning

1) Apply a small amount of solvent to a clean wiping cloth. Refer to 20-40-01, Silicone Adhesives – General Data Table 1, for the solvent appropriate for the surface to be cleaned. For solvents which evaporate quickly, apply more solvent than for solvents which evaporate slowly.

2) Wipe the area with the solvent dampened wiping cloth.

a) When cleaning bond surfaces with solvent saturate a clean lint-free cloth, and thoroughly scrub the bond surfaces. Wipe dry with a clean lint-free cloth before the solvent evaporates.

Note Wiping cloth (CID A–A–59323) shall be lint free, 100 percent cotton cloth conforming to A–A–2522, Grade A, colour 1: cotton gauze/cheese cloth conforming to CCC–C–440, Type II or III, which are unbleached or white cleaning cloths conforming to AMS 3819A, Class 2, Grade A, and have not been exposed to any other chemical solutions.

b) Keep solvent cloth saturated and turn frequently to avoid reusing a dirty area.

3) Wipe dry with another clean wiping cloth before the solvent evaporates. It is important to dry the surface immediately after cleaning because if the solvent evaporates before it is wiped up, oil and grease residue will be left on the cleaned surface. Although DS108 solvent has a slow evaporation rate, wipe dry as soon as possible as delay will result in a longer lingering odour.

4) Check for a water break free surface.

5) If more cleaning is needed, apply fresh solvent to a clean portion of the cloth and repeat Para 4.A.2), Para 4.A.3) and Para 4.A.4).

B. Mechanical Cleaning

1) Remove surface oxides or heavy deposits of contamination using aluminum wool or scotch-brite pad.

2) Dust off loose particles with a clean cloth.

C. Manual Deoxidizing (Deoxidine 624, Alumi Prep 33 or Turco W.O.1)

1) Apply manual deoxidizer using a cellulose sponge or sheepskin roller.

Note Close tolerance holes and surfaces must be masked before deoxidizing. Close tolerance holes do not require masking if final reaming of close tolerance holes will be done at the assembly stage.

Note Clean only small areas at a time. Do not allow the solution to dry on the surface. If necessary, apply the deoxidizer to a section of the part at a time.

2) Initial water rinse (must occur within 5 minutes of applying deoxidizer).

3) Wipe surface clean with water soaked cloths.

Note If the surfaces have been chemically conversion coated, wipe lightly to prevent damage to the coating.

Note Rinse cloths with clean water frequently.

4) Apply water moistened litmus paper while the surface is wet. If the litmus paper changes colour repeat Para 4.C.

5) If the component has laps or seams then apply water moistened litmus paper to the laps or seams. If litmus paper changes colour, blow out seams and laps with clean filtered compressed air. Repeat step until paper does not change colour.

If the component does not have laps or seams, if specified, manually apply chemical conversion coating in accordance to Para 6.

5. Preparation of Alodine 1200S Solution

WARNING WEAR RESPIRATOR TO PREVENT INHALATION OF ALODINE POWDER.

A. Prepare Alodine 1200S solution as follows:

1) Fill an acid resistant mixing container with clean luke warm water.

2) Carefully add the Alodine 1200S powder 1.0 to 3.0 wt oz per US gallon.

3) Thoroughly stir the solution until the powder is dissolved. Disregard the small amount of insoluble material which may settle out of solution.

B. Use Touch-N-Prep Coating Alodine 1132 marker as supplied.

6. Manual Application of Alodine 1200S Solution

A. Application of Alodine 1200S Chemical Conversion Coating

CAUTION TAKE EXTREME CARE WHEN PROCESSING ASSEMBLIES CONTAINING METALS OTHER THAN ALUMINUM. THE SOLUTIONS USED MAY BE CORROSIVE TO THESE METALS.

1) Apply chemical conversion coating to the surfaces of cleaned parts as follows:

a) For aluminum parts, apply conversion coating solution to the surface using a brush. Maintain a continuous wet surface.

b) Leave the solution on the part for 1 to 5 minutes. Do not allow the solution to dry on the surface. If drying does occur, re-wet the surfaces with solution before water rinsing.

c) Gently wipe all surfaces with a clean cloth soaked in the conversion coating solution just before rinsing.

d) Rinse parts thoroughly with water not exceeding a conductivity limit of 600 µmhos and pH of 5.0 to 8.0. Chemical salts trapped under a paint film will eventually result in blistering or corrosion problems.

e) Wipe all part surfaces dry using a clean lint-free cloth. Use clean oil free compressed air to blow dry treated surfaces such as lap joints, seams, pockets, etc.

f) Carefully inspect all treated surfaces and if necessary, re-apply the solution to bare or starved areas according to Para 6.A.1).a) through Para 6.A.1).e) in order to get the desired coating thickness. The coating colour ranges from light gold to tan depending on the metal type, temperature, solution concentration and number of applications. Ensure that the coating is not powdery.

g) For aluminum parts, prime parts as soon as possible after drying.

B. Touch Up of Chemical Conversion Coating Using Touch-N-Prep Alodine 1132 or Alodine 1200S

1) If using a Touch-N-Prep Coatings Alodine 1132 marker, touch-up parts as follows:

a) Remove cap.

b) If necessary, the applicator tip can be cut to conform to any shape with a single edge razor blade.

c) Hold unit with applicator tip down.

d) To activate, press tip against a firm surface. This will open the valve allowing Alodine 1132 to reach the applicator tip. A new unit should charge in 30 – 45 seconds. When the Alodine wets the tip, release pressure. Unit is ready to use.

e) Check to ensure that the surface to be coated is dry.

f) Apply Touch-N-Prep Coatings Alodine 1132 to the repaired surface in smooth, even strokes. Be sure to cover edges. For best results, the initial coating should be applied in one direction using a 50% overlap on each pass. After allowing the first coat to dry, apply the second coat at right angles to the first coat with 50% overlap. Do not allow the solution to puddle (puddling may result in improper drying leading to paint adhesion failures, flaking, etc.). A pre-warmed surface will result in a smoother coating. To recharge the applicator tip, repeat Para 6.B.1).d). Replace cap when not in use.

g) Allow the chemical conversion coating to air dry, force dry with warm air (eg. hair dryer) or using a heat lamp (130°F maximum). Do not rinse.

2) Except as specified in Para 6.B.1), touch-up parts with small surface areas with chemical conversion coating solution as follows:

a) Apply chemical conversion coating solution to prepared surface using a pencil brush, small Scotch-Brite pad or foam tip applicator to fully cover the applicable area. Prevent excess solution from dripping onto the surrounding structure.

b) Allow the solution to remain on the part surface for 1 to 5 minutes before washing off. Do not allow the solution to dry on the surface. If drying does occur, re-wet the surface with solution before water rinsing.

c) Remove all traces of solution by rinsing with clean water or wiping with water-damp cloths followed by wiping dry with clean lint-free cloth or blowing dry with clean oil-free compressed air.

C. Equipment Notes

1) Containers to hold the Alodine 1200S solution must be constructed from stainless steel, plastic, rubber or other acid proof material. Lead, glass, tin or galvanized iron must not be used.

2) Storing the solution in mild steel containers will result in a slow decomposition of the fluid.

7. Requirements

A. The treated surfaces must be free from bare or starved areas.

B. The coating must be continuous and free from powder, imperfections and excessive blotches which may result in poor paint adhesion.

C. The coating must be light gold to tan in colour.

8. Additional Information

A. Handling of Processed Parts

1) Do not touch conversion coated surfaces with bare hands. Wear clean cotton gloves when handling such parts.

2) For parts to be transported for further processing or to be held for more than 24 hours, wrap the parts completely or interlace with Kraft paper to ensure that the conversion coated surfaces are not exposed to contamination. Coated parts that are to be primed within 24 hours do not require protective wrapping but must be kept free from contamination and be handled only while wearing clean cotton gloves.

B. Defective Coatings

1) One or more of the following conditions may cause the conversion coating to be too light:

- The dwell time of alodine on the part was too short.
- The chemical concentration of the alodine solution is too low.
- The pH of the alodine solution is not within the specified limits.

2) One or more of the following conditions may be the cause of abnormal amounts of powder being encountered:

- The parts were improperly cleaned.
- The chemical concentration of the alodine solution is too high.
- The dwell time of the alodine on the part was too long.
- The pH of the alodine solution is not within the specified limits.

9. Storage

A. Store Alodine 1200S in a cool dry area away from organic compounds or easily oxidizable materials. Alodine 1200S is an acidic oxidizing agent and is not affected by freezing.

B. Store Touch-N-Prep Coatings Alodine 1132 markers away from incompatible materials. Alodine 1132 contains chromate. Protect from freezing. Do not tamper with packaging and do not try to refill used markers.

10. Disposal of Solutions

A. Dispose of chemical conversion coating solutions according to MSDS and all applicable regional regulations.

B. Dispose of used Touch-N-Prep Coatings Alodine 1132 markers by re-packaging used markers and mail back to supplier for disposal (mailing label is supplied in original packaging).

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